P8109 Spring 2026 Homework #5

Due on April 30 at 11:59pm

1. From a lot with an unknown fraction of defective items p, we select a simple random sample of n elements. Suppose that p is a random variable with density

$$f(p) = \begin{cases} 1, & 0 \le p \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

   Find the posterior probability $P(p \le 0.1)$ given that 2 out of 30 items are defective. You may have to use the beta distribution

$$f(p \mid \alpha, \beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\alpha-1}(1-p)^{\beta-1}, \qquad 0 < p < 1,$$

   and a software package to calculate the posterior probability.

2. Suppose the result X of an IQ test is normally distributed with unknown mean θ and variance 100. If the prior distribution of θ is $N(100, 225)$ and the observed score is $x = 115$, show that:

   (a) the posterior distribution of θ is $N(110.39, 69.23)$;

   (b) the Bayes factor for testing

$$H_0 : \theta \le 100 \quad \text{vs.} \quad H_1 : \theta > 100$$

   is 0.12.

3. An experiment started with $n = 20$ rabbits. Let X be the number that survived and let p be the true probability of survival for each rabbit. Then $X \sim \text{Binomial}(20, p)$.

   Suppose that 15 rabbits survived and that p has a beta prior with parameters $(\alpha, \beta) = (3, 5)$, i.e.,

$$f(p \mid 3, 5) = \frac{\Gamma(8)}{\Gamma(3)\Gamma(5)} p^2(1-p)^4, \qquad 0 < p < 1.$$

   Give your answers to two decimal places.

   (a) Compute the posterior Bayes estimator of p.

   (b) Obtain the maximum likelihood estimator of p.

   (c) How would your answer in (a) change if the prior of p were uniform on $(0, 1)$?

(d) Suppose now that 150 rabbits survived out of 200. Compare the MLE of p, the posterior Bayes estimator under a Beta$(3, 5)$ prior, and the posterior Bayes estimator under a uniform prior.

4. Let the conditional density of X given θ be gamma:

$$f(x \mid \theta) = \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-x/\theta}, \qquad x > 0,\ \alpha > 0,\ \theta > 0,$$

and let the prior density of θ be

$$g(\theta) = \frac{1}{\Gamma(\beta)\gamma^\beta} \theta^{\beta-1} e^{-\theta/\gamma}, \qquad \theta > 0,\ \beta > 0,\ \gamma > 0.$$

That is,

$$X \mid \theta \sim \text{Gamma}(\alpha, 1/\theta), \qquad \theta \sim \text{Gamma}(\beta, 1/\gamma).$$

Obtain the unconditional density of X.

5. Consider a parameter θ with prior density

$$\pi(\theta) = \gamma\beta^\gamma\theta^{-(\gamma+1)}, \qquad \theta \geq \beta,$$

where $\beta > 0$ and $\gamma > 2$.

(a) Find the prior mean of θ.

(b) Given observations $Y_1 = y_1, \ldots, Y_n = y_n$ from a Uniform$(0, \theta)$ distribution, find the posterior mean of θ.

(c) Show that the estimator

$$\hat{\theta} = \max\left\{z, \frac{\gamma + n}{\gamma + n - 1}\beta\right\}, \qquad z = \max(y_1, \ldots, y_n),$$

is admissible under squared error loss.

6. It is often claimed that traveling by plane is safer than traveling by car since the respective probabilities of a crash are approximately 5×10^{-6} and 10^{-2}. Even if these probabilities are accurate, is an airplane really much safer than a car?

	Travel by plane	Travel by car
Crash	10^6	100
No crash	5	0

Which decision is associated with a smaller risk?

7. Consider the following table of risks. The parameter space is $\Theta = \{\theta_1, \theta_2, \theta_3, \theta_4\}$ with prior probabilities $\{1/8, 3/8, 1/4, 1/4\}$. The decision space is $\{d_1, d_2, d_3\}$.

	d_1	d_2	d_3
θ_1	0	2	3
θ_2	1	0	2
θ_3	3	4	0
θ_4	1	2	0

(a) Find the Bayes risk of each decision.

(b) Which decision is Bayes?

(c) Obtain the minimax decisions.

8. Consider a single observation $X \sim \text{Bernoulli}(\theta)$. Suppose we estimate θ using a Bayesian estimator $d(X)$ such that

$$d(1) = d_1, \qquad d(0) = d_2,$$

with loss function

$$L(d, \theta) = (d - \theta)^2.$$

(a) Write down the risk function $R(d, \theta)$.

(b) Show that the minimax decision satisfies

$$d_1 = \frac{3}{4}, \qquad d_2 = \frac{1}{4}.$$